

EigenEngine

$$\hat{E}_i|\text{gen}\rangle \otimes \hat{E}_n|\text{gen}\rangle$$

Persistence and generation as problems of representation

EigenEngine, with Gerard McCaul

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The same structure as the companion paper, projected into a second form.

Two gaps

A language model returning to a growing body of work fails twice.

- **Persistence.** No state survives a session. The history is re-supplied each time, the window is bounded, and recall sags in the middle of a long input.
- **Generation.** With no record of its own past, output drifts toward the average of its training rather than toward this work.

Prompting, fine-tuning, reinforcement all steer a fixed model. None adds the missing piece: a structured store that persists and grows.

One problem, wearing two costumes

Both gaps are problems of representation. The same meaning rides in language or in vectors, and a question expensive in one is cheap in the other.

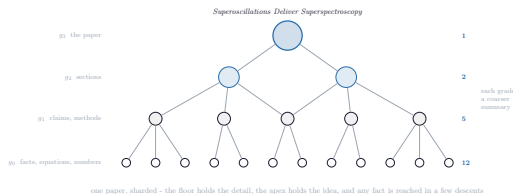
- Whether one concept contains another: an argument in prose, an inner product in vectors.
- Whether two records are the same person: an essay in prose, a membership test in a set.

So do not train a map from text to weights and read structure back out. Build the structure directly, and read it.

Grow the representation backwards

Fix the structure first; pay for intelligence once, when a thing is understood; let every later recall be arithmetic.

- Each document is placed in a cluster, each cluster under a theme, at ingest.
- The top layer holds one summary node per theme, not one per document.



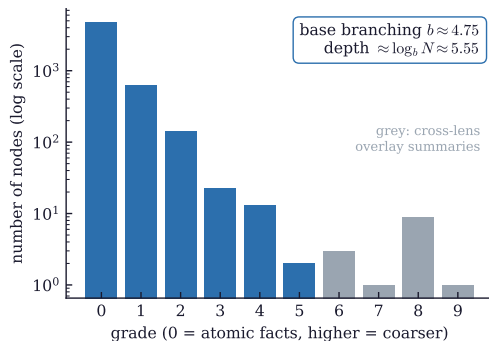
Recall is a short descent

Counting the live structure by grade:
4,829 $\rightarrow$ **637** $\rightarrow$ **144** $\rightarrow$ **23**.

A geometric decay, branching factor
 $b \approx 4.75$, so the depth to reach any of
5,662 nodes is

$$d \approx \log_b N \approx 5.5.$$

Intelligence is spent once in the build; the
query is the cheap walk down.

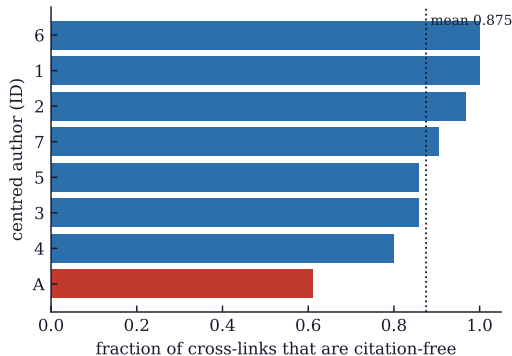


A query no citation graph can answer

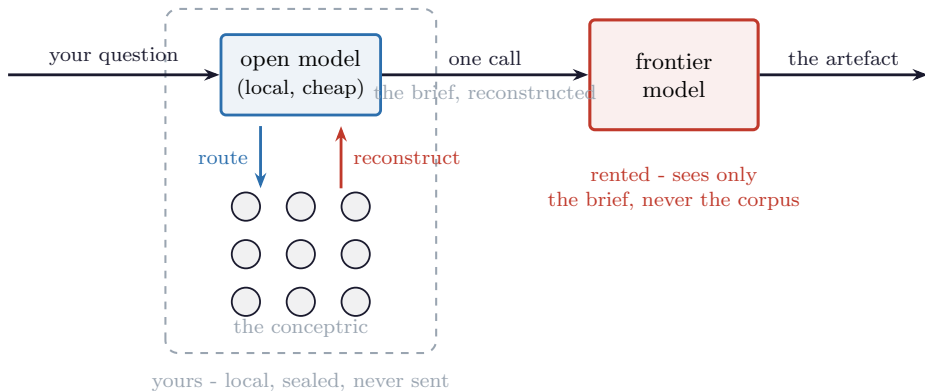
Cross-link one researcher's themes to every collaborator:

- **0** model inferences.
- **69** theme nodes touched; **1,307** papers never read.
- **11 of 18** links are citation-free, between researchers who share no paper.

Robust across all eight centres (mean 0.875); lowest where real co-authorships exist and are correctly recovered.



Generation is recall in reverse



A small local model walks the question down the structure and hands a frontier model one short brief. The frontier model never sees the corpus; intelligence is rented by the sentence.

Why not just retrieval-augmented generation?

A typical RAG or agent harness re-embeds documents and spends a model call on *every* query, over a flat, stateless index, retrieving existing passages by surface similarity.

- Here the intelligence is paid **once**, at fold time.
- Structural queries run by arithmetic over a small top: **zero** model calls, cost **logarithmic** in corpus size.
- One **persistent** substrate, reused across sessions.
- It surfaces relations that appear in **no single document**, which retrieval of existing text cannot find.

What is closed, and what is not

- **Closed:** persistence. The graded structure is built and queried cheaply on one researcher's machine.
- **Open:** voice and reasoning are still a general-purpose model wearing instructions. Making them the system's own needs an open-weight model trained on the structure.
- **Open:** larger corpora and a self-hosted generator need compute beyond one person.

This talk and its companion paper were projected by the engine they describe, from its own memory.

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Full paper: *Persistence and generation as problems of representation*

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